Binh Minh Tran

Contact Information

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Education

2027 – 2031 **Doctor of Philosophy, Medical Science**
Computational RNA Biology, John Curtin School of Medical Research, The Australian National University, Canberra, Australia.
PhD supervision and funding confirmed with Professor Eduardo Eyras.

Feb 2023 – Present **Bachelor of Advanced Computing (Honours)**
Computational Data Science, The University of Sydney, Australia.
Expected completion: December 2026. Cumulative WAM: **82/100 (Distinction)**.
Thesis: *Multimodal Breast Cancer Survival Prediction with Hierarchical Niche–Interface Graph Learning.*
Supervisor: Associate Professor Xiuying Wang.

2019 – 2022 **High School Diploma, Specialised Physics Programme**
HUS High School for Gifted Students, VNU University of Science, Vietnam National University, Hanoi, Vietnam.
GPA: **9.5/10**. Outstanding academic achievement award, 2019–2022.

Research Interests

Computational RNA Biology **Graph and Multimodal Learning** **Spatial Omics**
Computational Oncology **AI for Molecular Design**

Publications

Applicant's name in bold. † Equal contribution.

Published Peer-Reviewed Papers

[C1] **B. M. Tran**†, V. A. Nguyen†, L. H. Nguyen, Q. A. Dang, N. M. H. Nguyen, Q. A. Nguyen, H. N. Tran, T. B. V. Nguyen, and D. H. Tran, "A Multimodal Deep Learning Framework for Breast Cancer Receptor Status Prediction," *IUKM 2026*, LNAI 16918, Part I, Springer, pp. 102–114, 2026. DOI | Full paper.

[C2] L. H. Nguyen, **B. M. Tran**, and D. H. Tran, "A Variational Graph Neural Network with Multi-view Integration for Predicting lncRNA–Disease Associations," *Recent Challenges in Intelligent Information and Database Systems (ACIIDS 2026)*, CCIS 2964, Springer, pp. 355–369, 2026. DOI.

[C3] L. H. Nguyen, **B. M. Tran**, and D. H. Tran, "Dual-Stream Contrastive Graph Representation Learning for lncRNA–Disease Association Prediction," *Proceedings of the FISU Joint Conference on Artificial Intelligence (FJCAI 2026)*, pp. 151–158, 2026. ISBN 978-604-45-2586-0.

Manuscripts Under Review

[M1] **B. M. Tran**†, V. A. Nguyen†, Q. A. Dang, N. M. H. Nguyen, Q. A. Nguyen, H. N. Tran, D. H. Tran, and T. B. V. Nguyen, "Development and cross-cohort evaluation of a multimodal deep-learning model for ER, PR and HER2 status classification in breast cancer," *under review* at *Scientific Reports*, 2026.

[M2] **B. M. Tran**, D. Q.-A. Nguyen, M.-H. Ha, D. M. Pham, T. D. Nguyen, Q. B. Doan, D. T. Doan, M.-A. Nguyen, D.-T. Nguyen, and O. T.-C. Chen, "H2F-GCN: Skeleton-based Hypergraph Frequency Coordinate Fusion Graph Convolution Network for Physical Rehabilitation Assessment," *under review* at *IEEE Access*, 2026.

[M3] T. B. V. Nguyen, N. M. H. Nguyen, **B. M. Tran**, Q. A. Dang, and V. A. Nguyen, "A Deep Learning Model Supporting Classification and Localisation of Breast Tumour Lesions on Ultrasound" (in Vietnamese), *under review* at *Journal of Medical Research* (Hanoi Medical University), 2026.

[M4] **B. M. Tran**, L. Zhang, and X. Wang, "HiNT-Surv: Hierarchical Niche-Aware Multi-Microenvironment Learning from Imaging Mass Cytometry for Survival Prediction," *under review* for the *IEEE Symposium on Computational Intelligence in Health and Medicine (CIHM), SSCI 2027*.

Research Experience

Oct 2026 – Present

Visiting Researcher,
Centre for Computational Biomedical Sciences
John Curtin School of Medical Research, The Australian National University
Research host: Professor Eduardo Eyras.

- Undertaking preliminary research in computational RNA biology with Professor Eduardo Eyras in preparation for doctoral study.
- Research focus: AI/ML-based interpretation of genetic variants affecting RNA modifications (m^6A, pseudouridine and m^5C), distinguishing predicted modification changes from disease-relevant functional effects.

Feb 2026 – Oct 2026

Honours Research Student, School of Computer Science,
The University of Sydney, Australia. Supervisor: A/Prof Xiuying Wang.

- Designed NIMSurv, a hierarchical graph model integrating clinical data and imaging mass cytometry for breast-cancer survival prediction, with interpretable contributions from tissue interfaces.
- On 544 METABRIC patients, achieved 10-year disease-specific survival Uno's C of **0.703** under patient-level 5×5-fold cross-validation (+0.078 versus a clinical-plus-summary baseline).
- Exploratory transfer of a frozen common-input model to Basel yielded 5-year Uno's C of 0.725. An earlier model, HiNT-Surv, is under review [M4].

Dec 2025 – Present

Research Collaborator, Department of Obstetrics & Gynaecology,
Hanoi Medical University, Vietnam.

- Developed and implemented a multimodal, multitask model integrating clinical variables, H&E whole-slide images and tumour RNA-seq to predict ER, PR and HER2 status in breast cancer [C1].
- Conducted patient-level stratified five-fold cross-validation on 937 TCGA-BRCA cases (macro AUC **0.890**) and frozen external testing on 122 CPTAC-BRCA cases (macro AUC **0.931**; no retraining, tuning or threshold adjustment).

Dec 2025 – Mar 2026

Research Assistant, Center for Biomedical and Pharmaceutical Research, Vietnam Academy of Science and Technology (VAST), Vietnam.

- Applied reproducible NGS preprocessing and annotation workflows, including FastQC, Trimmomatic, genome assembly, gene prediction and functional annotation, to mutation and microbial gene-function analyses.

Dec 2024 – Dec 2025

Research Collaborator, Hanoi University of Industry, Vietnam.
Supervisor: A/Prof Dang Hung Tran.

- Developed and implemented two graph-learning models for lncRNA–disease association prediction: a variational graph neural network with multi-view integration [C2] and a dual-stream contrastive graph model [C3].
- Evaluated both models and drafted the manuscripts, contributing to two peer-reviewed conference publications.

Funded Research Project

Dec 2026 – Nov 2027

AI Model for Individualised Post-Test Probability after High-Risk NIPT
Department of Obstetrics & Gynaecology, Hanoi Medical University.

Role: Co-principal investigator with Nguyen Van An
Funding: **VND 10,000,000** awarded by Hanoi Medical University
Supervisor: Nguyen Thi Bich Van, MD, PhD

Aim: Develop and calibrate an individualised probability model for confirmed trisomy 21, 18 or 13 following a high-risk NIPT result, using signed Z-score, fetal fraction and assay platform in retrospective high-risk referrals.

Scholarships

2027 (prospective)

MACSYS PhD Scholarship
ARC Centre of Excellence for the Mathematical Analysis of Cellular Systems (MACSYS), The Australian National University.
Supports doctoral research in Medical Science (Computational RNA Biology) at the Centre for Computational Biomedical Sciences, John Curtin School of Medical Research.
Funding confirmed in writing by Professor Eduardo Eyras, subject to formal PhD admission.

2023 – Present

Sydney International Student Award, The University of Sydney — tuition support awarded at admission for the duration of the degree.

Awards

2026

Selected Top 10 Finalist, AURORA Scientific Research Awards, Vietnam–Australia Innovation Network (NIC-AU). Grand Final scheduled for December 2026, Hanoi.

2026

Outstanding Student Award, Embassy of Vietnam in Australia — academic and research achievement.

2026

Second Prize, Vietnamese Young Intellectuals Forum in Australia — breast-cancer receptor-status research [C1].

2026

Third Prize, Youth Science and Technology Conference, Hanoi Medical University [C1]; selected for the 23rd Youth Science and Technology Conference of the Vietnamese Health Sector.

2025

January Star Award, Vietnam National Union of Students — academic achievement, research and student leadership; one of 117 national recipients.

Industry Experience

Dec 2024 – Mar 2025	**Data Engineer Intern**, Viettel Software, Hanoi, Vietnam. Built and maintained Python/SQL ETL pipelines; contributed to data-quality monitoring, schema design and pipeline/query performance tuning.
Jun – Jul 2024	**Data Engineer Intern**, Agribank, Hanoi, Vietnam. Supported ETL into reporting systems and contributed to dashboards and operational reports for internal stakeholders.

Technical Skills

Programming	Python, R, SQL, Bash; Git, Linux, LaTeX, experiment tracking.
Frameworks	PyTorch, PyTorch Geometric, scikit-learn.
Methods	Graph neural networks; multiple-instance, self-supervised and contrastive learning; multimodal fusion; Cox and discrete-time survival models.
Evaluation	Patient-level nested cross-validation; bootstrap confidence intervals; paired tests with multiple-testing correction; effect sizes; calibration; external validation.
Biomedical data	Imaging mass cytometry; RNA-seq and gene expression; H&E whole-slide images; NGS preprocessing and annotation; lncRNA–disease networks.
Languages	Vietnamese (native); English (IELTS 7.0); French (DELF A1).

References

Referee 1	**Associate Professor Xiuying Wang** School of Computer Science, The University of Sydney. Email: xiu.wang@sydney.edu.au
Referee 2	**Associate Professor Dang Hung Tran** Hanoi University of Industry. Email: hungtd@haui.edu.vn
Referee 3	**Professor Eduardo Eyras** John Curtin School of Medical Research, ANU. Email: eduardo.eyras@anu.edu.au